What's New in PalmQ10 v0.1.3-alpha

Released 19 July 2026 • Private research release

1 At a glance

The headline change is that **PalmQ10 can now run simulations on an HPC cluster**, and connecting to one no longer means typing your cluster password repeatedly. You enter a single master password when a job starts, and everything after that — uploads, job submission, monitoring, downloading results — happens without further prompts, surviving VPN and WiFi interruptions.

Alongside that, this release carries a set of correctness fixes in the simulation chain itself. Several affected results rather than convenience: a vertical datum mismatch, a post-processing extraction height error, and a spin-up setting that ignored what you chose in the interface.

If you are upgrading from v0.1.1-alpha

You almost certainly are. Version v0.1.2-alpha was tagged but shipped without an installer or the geographic dataset, so it was never installable — its assets were downloaded zero times. Everything in Section 3 therefore applies to you as well.

2 Changes since v0.1.2-alpha

2.1 HPC cluster authentication

This is entirely new. Previously each stage of a cluster job authenticated separately, so a single run prompted for your cluster password four or five times — and any brief network interruption dropped the connection and prompted again.

- **One master password per job.** You choose a master password once. It is local to your machine, is never stored or transmitted, and unlocks an encrypted vault holding the key PalmQ10 uses to log in for you.
- **Your cluster password is entered once, ever.** During first-time setup it is used a single time to register PalmQ10's key with the cluster, then discarded. It is never written to disk.
- **Silent reconnection.** Because login is key-based, a dropped VPN or WiFi connection reconnects invisibly, and interrupted file transfers resume where they stopped rather than restarting.
- **Master password management in the GUI.** A dedicated button creates or changes your master password. The vault is shared across clusters, so a second cluster reuses the same master password.
- **Enabled by default** on the supplied Rigel and Sirius profiles. Profiles can opt out with `auth_mode: password_interactive`, which restores the previous behaviour unchanged.

Three separate code paths previously bypassed this and prompted for credentials anyway — the WRF4PALM pre-flight scratch check, the HPC asset upload buttons, and the WPS_GEOG transfer. All three now authenticate up front and share a single unlocked session.

2.2 Large transfers over unreliable connections

Uploading a container image or the geographic dataset to a cluster can take hours, which made them vulnerable to VPN sessions that time out overnight.

- Asset uploads now retry **indefinitely** rather than giving up after a few attempts, reporting progress as they go, and resume partial transfers.
- Job input uploads keep a bounded retry count, so genuine errors still surface instead of hanging a run.

2.3 Correctness fixes in the simulation chain

Area	Fix
Vertical datum	The PALM domain floor and the atmospheric forcing derived from WRF used inconsistent reference heights, placing the atmosphere roughly 200 m too low for elevated terrain. Origin elevation is now set from the minimum terrain height in your domain and applied consistently to the configuration and the namelist.
Thermal comfort outputs	Post-processing extracted horizontal fields at a level that ignored the domain origin elevation, sampling roughly 200 m too high — which produced zero radiant temperature and, in turn, empty comfort results. Mean radiant temperature was additionally sampled at a fixed height rather than its own near-surface level. Both are corrected.
Spin-up duration	The spin-up setting in the interface was ignored and a fixed six hours was always used. Your chosen value is now honoured.
Building integration	Improvements to building extraction, geometry repair, and PALM surface-type assignment.
Terrain sources	Updates to the FABDEM, ESA/COP30, and NASA elevation download paths, and to canopy height preparation.
WRF and WRF4PALM	Updates to namelist generation, ERA5 retrieval, and the WRF4PALM driver build.

2.4 Documentation

This release adds a User Guide, an Installer Guide, a Cluster Connection Manual, a Technical Guide, and this document. Earlier releases shipped installer notes only.

3 Also new if you are coming from v0.1.1-alpha

Version v0.1.2-alpha introduced the following. Since it was never installable, these will be new to you as well.

- **HPC execution itself.** v0.1.1 ran locally only. The scheduler and storage adapters that submit and monitor cluster jobs — SLURM and SGE, Lustre storage — arrived in v0.1.2.

- **Cluster profiles** describing a cluster in YAML, with supplied profiles for the UPV Rigel and Sirius systems.
- **Computing backend selector** in the interface, switching between local and HPC execution.
- **Vertical stretching control** in the interface. Note that the feature itself is **not yet functional** — see Known limitations below.
- **Updated QGIS plugin** with the cluster profile dialog and revised layout.

4 Unchanged in this release

- **Container images.** All five Apptainer images are byte-for-byte identical to those shipped in v0.1.1 and v0.1.2. If you already have them, they have not changed.
- **Conda environment.** The `palmq10.yml` environment specification is unchanged.
- **Geographic dataset.** The WPS_GEOG archives are unchanged from v0.1.1.

5 Known limitations

Building data covers Valencia and Lecco only

Building footprints and heights come from a prebuilt dataset covering **Valencia (Spain)** and **Lecco (Italy)** — the two areas currently provisioned for `eubds` access. The interface presents `eubds` as a building source, but it is an extract for those two cities, not a general European database. An AOI elsewhere will simulate without realistic buildings. Terrain, land cover, streets, and weather forcing are global; only buildings are restricted.

- **Vertical stretching is not functional.** The *Vertical stretching* control appears in the interface, but the feature is not yet implemented. Leave it set to `Off`; it is planned for a future release.
- **Windows only** — Windows 10/11 with QGIS and Ubuntu on WSL2.
- **Lake physics is not enabled.** Water bodies use a fixed temperature. Lake depth data and the associated WRF physics are planned for a future release.
- **Cluster support is validated on two systems** (UPV Rigel and Sirius). Other SLURM or SGE clusters should work but are untested.
- **Alpha status.** Some steps require user interaction, and some failures surface as raw model errors.

6 Upgrade notes

1. Run the v0.1.3 installer as described in the Installer Guide. A GitHub token is required, as before.
2. **Re-fill your access keys.** `ACCESS_KEYS_EDIT.txt` ships blank. Copy your values across from your previous installation rather than re-registering with each provider.
3. **Set a master password** before your first cluster job. See the Cluster Connection Manual.

4. The container images have not changed. If you are reinstalling on a machine that already has them, they do not need downloading again.
